Haojie Ye

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EDUCATION

Ph.D. in Computer Science and Engineering June 2021 – Sept 2024
Thesis Advisor: Prof. Trevor Mudge University of Michigan, Ann Arbor

- **Overall GPA**: 4.0/4.0
- **Area Focus**: Computer architecture and hardware-software co-design for recommendation inference, graph processing, and memory systems

M.S.E. in Computer Science and Engineering Sept 2019 – May 2021
Thesis Advisor: Prof. Trevor Mudge University of Michigan, Ann Arbor

- **Overall GPA**: 4.0/4.0

B.S.E. in Computer Engineering Sept 2017 – May 2019
University of Michigan, Ann Arbor

FULL-TIME POSITION

NVIDIA — Deep Learning Performance Architect Oct 2024 – Present
Performance Modeling and Architecture Pathfinding Santa Clara, California

- Led LLM and video-generation performance studies across **Blackwell** and **Rubin** architectures, including **RTX PRO 6000 Blackwell**, to guide architecture and product planning.
- Developed and validated models for **LLM prefill and decode** and **diffusion-transformer video generation**; analyzed sparse attention and **tensor, expert, and context parallelism** to explain compute, memory, and communication bottlenecks.
- Independently built an **AI-agent-assisted performance-attribution component** that reconciles execution mappings, explains differences between projections, and flags outliers for repeatable comparison and targeted debugging.
- Evaluated **FP8/FP4 precision**, memory capacity and bandwidth, and interconnect tradeoffs; controlled for execution-mapping differences when interpreting end-to-end results and forming product recommendations.
- Presented technical findings to architecture and product teams, developed study materials, and taught colleagues how to run and explain GPU performance studies.

SELECTED PUBLICATIONS

1. **H. Ye**, Y. Xia, Y. Chen, K.-Y. Chen, Y. Yuan, S. Deng, B. Kasikci, T. Mudge, N. Talati, "Palermo: Improving the Performance of Oblivious Memory using Protocol-Hardware Co-Design" (**HPCA'25**)
2. Y. Xia, J. Kim, Y. Chen, **H. Ye**, S. Kundu, C. Hao, N. Talati, "Understanding the Performance and Estimating the Cost of LLM Fine-Tuning" (**IISWC'24**)
3. Y. Chen, **H. Ye**, S. Vedula, A. Bronstein, R. Dreslinski, T. Mudge, N. Talati, "Demystifying Graph Sparsification Algorithms in Graph Properties Preservation" (**PVLDB 2023 / VLDB'24**)
4. Y. Yuan, **H. Ye**, S. Vedula, W. Kaza, N. Talati, "Everest: GPU-Accelerated System For Mining Temporal Motifs" (**PVLDB 2023 / VLDB'24**)
5. **H. Ye**, S. Vedula, Y. Chen, Y. Yang, A. Bronstein, R. Dreslinski, T. Mudge, N. Talati, "GRACE: A Scalable Graph-Based Approach to Accelerating Recommendation Model Inference" (**ASPLOS'23**)
6. H. Kim, **H. Ye**, T. Mudge, R. Dreslinski, N. Talati, "RecPIM: A PIM-Enabled DRAM-RRAM Hybrid Memory System For Recommendation Models" (**ISLPED'23**)

7. N. Talati, **H. Ye**, S. Vedula, K.-Y. Chen, Y. Chen, D. Liu, Y. Yuan, D. Blaauw, A. Bronstein, T. Mudge, R. Dreslinski, "Mint: An Accelerator For Mining Temporal Motifs" (**MICRO'22**)
8. L. Belayneh, **H. Ye**, K.-Y. Chen, D. Blaauw, T. Mudge, R. Dreslinski, N. Talati, "Locality-aware Optimizations for Improving Remote Memory Latency in Multi-GPU Systems" (**PACT'22**)
9. N. Talati, **H. Ye**, Y. Yang, L. Belayneh, K.-Y. Chen, D. Blaauw, T. Mudge, R. Dreslinski, "NDMiner: Accelerating Graph Pattern Mining Using Near Data Processing" (**ISCA'22**)
10. N. Talati, D. Jin, **H. Ye**, A. Brahmakshatriya, G. Dasika, S. Amarasinghe, T. Mudge, D. Koutra, R. Dreslinski, "A Deep Dive Into Understanding The Random Walk-Based Temporal Graph Learning" (**IISWC'21**)
11. X. He, S. Pal, A. Amarnath, S. Feng, D.-H. Park, A. Rovinski, **H. Ye**, Y. Chen, R. Dreslinski, T. Mudge, "Sparse-TPU: Adapting Systolic Arrays for Sparse Matrices" (**ICS'20**)

INTERNSHIP

NVIDIA — Deep Learning Performance Architect Intern May 2023 – Aug 2023
Performance Modeling and Projection Santa Clara, California

- Researched LLM performance modeling and projection methodologies for future GPU architectures.

Micron Technology — Advanced Hardware Development Intern May 2022 – Aug 2022
In-memory computing research Allen, Texas

- Researched enhanced in-memory computing functions and memory-system design.

TECHNICAL SKILLS

- **Programming Languages**: C/C++, Python, CUDA, Verilog HDL, SystemVerilog, HTML, MATLAB
- **Software and Design Tools**: Git, Murphi, Cadence, Design Compiler, LaTeX
- **Architecture Simulators**: gem5, gem5-gpu, DRAMSim, Ramulator, CACTI, zsim

TEACHING EXPERIENCE

Graduate Student Instructor Sept 2020 – Jan 2021
EECS 598 Applied Parallel Programming with GPUs University of Michigan, Ann Arbor

- Taught with Prof. Reetuparna Das; prepared lecture slides, homework, and exams on GPU microarchitecture and CUDA programming.

AWARDS & HONORS

- Outstanding Undergraduate Research Award (2019)
- James B. Angell Scholar (2019)

REFERENCES

- Prof. Trevor Mudge, University of Michigan — tnm@umich.edu
- Prof. Ronald Dreslinski, University of Michigan — rdreslin@umich.edu
- Dr. Nishil Talati — talatin@umich.edu